

A Formal Semantic Analysis of Plurality and Definiteness in Modern Colloquial Persian

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Abstract

Persian can form plurals by combining a noun (e.g. *ketab* “book”) with the plural morpheme (e.g. *ketab-ba* “books”) or alternatively numerals and classifiers, without the plural morpheme (e.g. *do/se/tjænd ta kitab* “two/three/some books”). The first is often interpreted as definite and the second as indefinite. Previous research suggests that the plural morpheme *-ba* makes nouns definite, while the numerals *jek/do/.../tjænd* make them indefinite. This study argues against this view by looking at five plural constructions in Persian: bare plurals, indefinite plurals, classifier plurals, negative plurals, and plural constituent questions. It presents evidence suggesting that despite taking the plural morpheme, bare plurals and indefinite plurals receive indefinite interpretations systematically. Therefore, the plural morpheme is unlikely to encode for definiteness. It shows that the plurals with numerals and classifiers can be definite. Therefore, it is unlikely that numerals encode for indefiniteness. It also shows that in both singular and plural constructions, definiteness is the result of a variety of factors including the presence/absence of the indefinite clitic, prosody, background knowledge, tense, and aspect. The paper presents a formal semantic analysis of Persian plurals in which plurality and definiteness are introduced via separate mechanisms: plurality is introduced morphologically via the plural marker or numeral classifiers, and (in)definiteness is introduced via type-shifting operations that can be cued through different factors such as prosody, world-knowledge, or tense and aspect.

1 Introduction

This paper focuses on the semantics of plurality, definiteness, and their interaction in colloquial Persian. Languages vary in how they convey the number and plurality of their nominals (Chierchia 2021; Dryer 2013). Some languages like English can combine numerals directly with plural count nouns (e.g. *three apples*) but for mass nouns, they combine numerals with classifiers (measure phrases) and bare nouns (*three drops of blood*). Others like Mandarin Chinese always combine numerals and classifiers with bare nouns, whether the noun is conceptually count or mass. A third group of languages, like Turkish and Persian (Sağ-Parvardeh 2019), can apply both strategies as shown in (1).

- (1) Example bare plural and classifier plural constructions in Persian:

- a. *daneʃdʒu-ba* *hæf zæd-æn*
university.student-PL talk hit.PST-3.PL
‘The students talked.’

- b. *tʃænd ta danɛʃdʒu hæfzæd-æn*
 NUM CL uni.student talk hit.PST-3.PL
 ‘Some/a-few students talked.’

Previous research has suggested that the plural suffix *-ba* makes nouns both plural and definite (Gebhardt 2008, 2009; Ghaniabadi 2009; Ghomeshi 2003; Jasbi 2015; Mirrazi 2021). In other words, *-ba* is like the English plural morpheme *-s* and the definite article *the* fused together. When we observe typical Persian examples like (1a) the conclusion seems inevitable. The noun phrase *danɛʃdʒu-ba* is translated as ‘the students’, and since *danɛʃdʒu* means ‘student’, we may conclude that the plural and the definite implications both come from the morpheme *-ba*. However, there is also evidence like (2b) below suggesting that the plural marker can appear on noun phrases that receive an indefinite interpretation (Sağ 2024; Sağ-Parvardeh 2019). How can we reconcile these facts and provide a unified account for the semantics of plurality and definiteness in Persian?

(2) Definite and indefinite interpretation of plural nouns with *-ba*:

- a. *danɛʃdʒu-ba hæfzæd-æn*
 university.student-PL talk hit.PST-3.PL
 ‘The students talked.’
- b. *danɛʃdʒu-ba-je ziad-i hæfzæd-æn*
 uni.student-PL-EZ many-IND talk hit.PST-3.PL
 ‘Many students talked.’

In this paper, I first present an overview of definiteness in Persian singular nominals. I argue that in singular nominals, definiteness is conveyed by a variety of factors including the absence of indefinite morphemes, phrasal intonation, tense-aspect-mood (TAM), specificity markers, and world knowledge. Then I show that a similar pattern exists with plural constructions. I present a systematic investigation of five plural constructions in common colloquial Persian in Section 3, showing that the indefinite interpretations go beyond a limited number of exceptions. Then in Section 4, I follow this descriptive work with a formal analysis that decouples plurality from definiteness. In this analysis the plural morpheme *-ba* implies plurality only; definiteness is supplied by other factors such the absence of an indefinite morpheme, phrasal intonation, TAM, world knowledge, etc. Under this analysis, the same mechanisms provide definiteness for singular and plural nominals in colloquial Persian.

Before I start, it is important to establish common ground regarding some empirical and analytical choices. First, it is crucial that we carefully distinguish between examples and data that belong to the informal (colloquial) variety of Persian vs. the formal variety. While this issue is acknowledged by most linguists working on Persian, the degree to which these varieties differ is commonly underestimated. It is often assumed that the differences are mainly phonological - e.g. *xane* ‘house (formal)’ vs. *xune* ‘house (informal)’ — and therefore syntactic and semantic inquiry can continue using data from either variety without major risks to the derived generalizations. While in many cases the informal and formal variants of an utterance only differ phonologically, there are also many cases where these varieties differ in morphosyntax and semantics. Example (3) shows a formal sentence in (3a) and its informal variant in (3b), highlighting four functional and morphosyntactic differences.

(3) Example differences between informal and formal Persian:

- a. *tʃænd danɛʃdʒu dar ketabxane-i ketab-ha-i xand-ænd*
 NUM student in library-IND book-PL-IND read.PST-3.PL
 ‘Some students read some books in a library.’
- b. *tʃæn(d)-ta danɛʃdʒu tu je ketabxune je ketab-ha-i xund-æn*
 NUM-CL student in IND library one book-PL-IND read.PST-3.PL
 ‘Some students read some books in a library.’

The formal variety does not require the classifier *-ta* while the informal variety does. The appropriate preposition for ‘in’ is *dar* in the formal variety and *tu* in the informal variety. The indefinite clitic *-i* can make a noun indefinite by itself in the formal variety but not in the informal variety; the indefinite numeral *je(k)* must be used instead. The plural morpheme *-ha* and the indefinite clitic *-i* can create a plural indefinite in the formal variety much more easily than in the informal variety. These differences, among many others, suggest that we are dealing with two distinct but related linguistic systems and we need to be careful when making grammatical generalizations. Especially that only one variety, namely informal Persian, is learned naturally in childhood and is commonly used by speakers for everyday conversations. The formal variety is mainly taught in schools and through formal education, and is reserved for specific formal and literary contexts. Therefore, the informal variety is much more suitable for linguistic inquiry as a natural language than the formal one. This is why in this paper I focus on the informal variety of Persian and use the International Phonetic Alphabet to help readers hone their intuitions and judgments using the informal variety’s phonology.

Finally, for reliable generalizations and valid causal inference based on isolated linguistic examples, it is crucial to use sentences that are “minimal pairs”. In semantics and pragmatics, the minimal pairs should be structured such that a change in exactly one aspect of the utterance form (e.g. presence or absence of a morpheme) results in a change in the semantic/pragmatic implication of the utterance (e.g. definite vs. indefinite interpretations). As the number of factors under consideration increases (e.g. presence of a morpheme, different syntactic positions, prosody, etc.), the number of minimal-pair contrasts required to test the role of each factor increases as well. This is why the empirical data provided in this paper are often in the form of a relatively long list of examples with minimal variations. I understand that this is not the orthodox way of presenting data in linguistic papers and it might be confusing, but I believe this practice will help us have more accurate empirical generalizations and theoretical accounts.

2 Definiteness in Singular Nominals

Bare nouns in Persian can receive definite, number neutral indefinite, or generic interpretations (Mirrazi 2021). The examples in (4) show some of these interpretations for bare nouns in the subject, direct object, and PP object positions. The sentence in (4a) can be true if: 1. a unique child known to the speaker and hearer (i.e. definite) is playing in a unique street known to them (definite); 2. one or more children (number neutral indefinite) were playing in a unique street known to both speaker and hearer (definite); 3. one or more children (number neutral indefinite) were playing in one or more streets of the city (number neutral indefinite); or 4. playing in the streets is what children generally do. These readings can be better accessed if the utterance is the answer to one of the following questions under discussion (QUD) respectively: 1. what is the child doing in the street? 2. What is happening in the street? 3. What is going on in this city? 4. What does a child do generally? Each of these interpretations typically receives a different intonational contour.

(4) Example interpretations of Persian bare nouns:

- a. *bætʃe tu xiabun bazi mi-kon-e*
 child in street play IMP-do.NPST-3.SG
 ‘1. The child is playing in the street.
 2. Some (number of) child(ren) are playing in the street.
 3. Some (number of) child(ren) are playing in some (number of) street(s).
 4. Children (generally) play in streets.’
- b. *diruz bætʃe æz deræxt sib kænd*
 yesterday child from tree apple remove.NPST
 ‘1. Yesterday, the child picked some (number of) apple(s) from the tree.
 2. Yesterday, the child picked some (number of) apple(s) from some (number of) tree(s).’

The example in (4b) shows similar interpretations for the direct object and PP object positions. While the subject position bare noun is most likely interpreted as “the child” (definite), the PP object and the direct object are interpreted as numberless indefinite. The sentence is true if the child picked one or more apples from a unique tree known to the discourse participants or from one or more trees. The upshot is that bare nouns in Persian are under-specified and can receive definite, number neutral indefinite, or generic interpretations in various syntactic positions depending on several factors such as discourse context, prosody, tense, and aspect, etc. Some linguists have treated number neutral indefinite interpretations of bare nouns in Persian as special cases that rely on a specific syntactic position or construction. As far as I can see this is not the case. It is easy to find number neutral interpretations of bare nouns in almost any syntactic position as the examples above show. Therefore in my analysis later in the paper, I treat number neutrality of bare nouns as a main feature of their semantics and not a special case.

Persian has two morphemes that often appear on nouns with indefinite interpretations: the numeral *je(k)* “one” and the indefinite clitic *-i*. Colloquial Persian can use the numeral alone to make simple indefinites like “an apple” in (5a) but the same cannot be done with the indefinite clitic as (5b) shows. A noun with the indefinite clitic has a restricted distribution in colloquial Persian and often needs special environments to be acceptable.

- (5) a. *je sib ru miz e*
 one apple on table be.NPST.3.SG
 ‘An apple is on the table’
- b. * *sib-i ru miz e*
 apple-IND on table be.NPST.3.SG
 ‘An apple is on the table.’

The examples in (6) list a few environments in which the indefinite clitic *-i* is acceptable. The distribution of the indefinite clitic in Persian is in some ways similar to the distribution of *any* in English and to other negative polarity items in world languages. Jasbi (2016) provides a more thorough analysis of the indefinite clitic *-i* in Persian; also see Moghiseh & Alonso-Ovalle (2019a,b, 2022).

- (6) a. Epistemic Indefinite:
je sib-i ru miz e
 one apple-IND on table be.NPST.3.SG

- ‘An apple is on the table.’
- b. Modification:
sib-e bozo/g-i ru miz e
 apple-EZ big-IND on table be.NPST.3.SG
 ‘A big apple is on the table.’
- c. Negation:
sib-i ru miz ni-st
 apple-IND on table NEG-be.NPST.3.SG
 ‘No apple is on the table.’
- d. Question:
sib-i ru miz e?
 apple-IND on table be.NPST.3.SG
 ‘Is an apple on the table?’
- e. Conditional Antecedent:
age sib-i ru miz e, bær-ef dar!
 if apple-IND on table be.NPST.3.SG, pick-3.SG NPST.have
 ‘If an apple is on the table, pick it up!’

Previous studies have analyzed the numeral *je* as an indefinite determiner similar to *a(n)* in English (Jasbi 2016, 2020a,b). However, examples like (7) below show that such an analysis may be incorrect. In (7a), the noun “apple” is modified only by the demonstrative determiners but in (7b), the noun is modified by *je* in addition to the demonstrative determiners. The final construction can be interpreted as “this/that one apple”.

- (7) a. *in/un sib ru miz e*
 this/that apple on table be.NPST.3.SG
 ‘This/that apple is on the table.’
- b. *in/un je sib ru miz e*
 this/that one apple on table be.NPST.3.SG
 ‘This/that one apple is on the table.’

Compare the behavior of *je(k)* in (7) with the behavior of the English numeral *one* vs the indefinite determiner *a(n)* in (8). Of course a nominal with the English indefinite determiner like “an apple” cannot be further modified by the definite or demonstrative determiners as (8c) shows. The numeral “one”, on the other hand, has no issue combining with the definite and demonstrative determiners in (8d). In this respect, the behavior of *je(k)* is similar to the English numeral *one* and not the English indefinite determiner *a(n)*. Given that *je(k)* is part of the Persian number system, analyzing it as a numeral rather than an indefinite determiner makes more intuitive sense as well. This analysis has an important consequence for Persian because it will reclassify the language as one that has no definite or indefinite determiners.

- (8) a. An/one apple is on the table.
 b. This/that apple is on the table.

- c. * This/that/the an apple is on the table.
- d. This/that/the one apple is on the table.

Persian does not have any morpheme that can correspond to definite determiners like *the*. The nominal (specificity) suffix *-e* and the object marker *ra* shown in example (9a) come close but further investigations have shown that they can also receive indefinite interpretations as (9b) shows, and their semantic contributions correspond roughly to different flavors of specificity. See Jasbi (2020a,b) for a more detailed discussion of the semantics of these morphemes and their connection to definiteness.

- (9) Nominals with the specificity marker *-e* or object marker *ra* can have indefinite interpretations
- a. *mærd-e sib-o xord*
man-SP apple-ACC eat.PST.3.PL
'The man ate the apple.'
 - b. *je mærd-e je sib-o xord*
one man-SP one apple-ACC eat.PST.3.PL
'A (certain) man ate an apple.'

If Persian has no morpheme for definiteness, then how does it convey that a nominal is definite? The language has a variety of linguistic devices and strategies to convey definiteness depending on the context, including the absence of the indefinite numeral *je(k)* "one" or the indefinite clitic *-i* (Nomoto 2013), phrasal stress (Krifka & Modarresi 2016), tense-aspect-mood (TAM), case marking (i.e. the object marker *ra*), and specificity morphemes. The examples in (9) show how the absence of the indefinite numeral *je(k)* in (9a), along with the presence of a specificity morpheme and an object marker, allows for a definite interpretation while the presence of *je(k)* in (9b) creates indefinite interpretations for the subject and the object of the sentence.

Intonation and phrasal stress can also cause definite or indefinite interpretations of a bare nominal (Krifka & Modarresi 2016). In (10a), the bare noun *sib* "apple" is stressed compared to the verb phrase which can result in a number neutral indefinite interpretation: "There is some (amount/number of) apple(s) on the table." The intonation can be better elicited if we imagine it as a response to the question "what is on the table?". In (10b) however, the stress is on the verb phrase rather than the bare noun which results in a definite interpretation of the bare nominal. This interpretation can be construed as a response to the question: "Where is the apple?". The explicit presence of these QUDs are not necessary for the mentioned interpretations but they help readers better imagine the phrasal stress and intonation of the utterance.

- (10) a. QUD: What is on the table? [*tʃi ru miz e?*]
[*sib*]_f *ru miz e*
apple on table be.3.SG
'There's apple on the table.'
- b. QUD: Where is the apple? [*sib kodʒa-st?*]
sib [*ru miz*]_f *e*
apple on table be.3.SG
'The apple is on the table'

The tense-aspect-mood (TAM) system of Persian can also enforce definite interpretations of bare nominals. The examples in (11) show this by minimally changing the aspect and tense of the verb *hæf zæden* “talking”. In (11a) the verb is in the past tense form and the imperfective morpheme *mi-* is absent. As a result, the only available interpretation is one in which both bare nouns are interpreted as definite: “the professor talked to the student.” This reading can be further enhanced by adding the adverbial *diruz* “yesterday”. The habitual or generic interpretations are not available for (11a). If we add the adverbials *mæmulæn* “usually” or *kol:æn* “generally” to the utterance, the utterance becomes unacceptable¹. In (11b) I added the imperfective morpheme *mi-* to the verb and as a result we have the habitual and generic interpretations available. The bare nouns in (11b) can be interpreted as definite (“The professor was talking to the student”) or past habitual/generic (“Professors used to talk to students”). The utterance is acceptable with either of the adverbials *diruz* yesterday, *mæmulæn* “usually”, or *kol:æn* “generally”, although the adverb *kol:æn* “generally” is less suitable for such past tense statements and more suitable for non-past examples like (11c). The example in (11c) uses the non-past verb stem and can therefore be interpreted as present definite (“The professor talk to the student (today)”) or present habitual/generic (“Professors (usually/generally) talk to students.”).

(11) Examples for the interaction of tense and aspect with definiteness

- a. (*diruz/#mæmulæn/?kol:æn*) *ostad ba danešdzu hæf zæd*
yesterday/usually/generally professor with uni.student talk **hit.PST**
‘The professor talked to the student (yesterday).’
- b. (*diruz/mæmulæn/?kol:æn*) *ostad ba danešdzu hæf mi-zæd*
yesterday/usually/generally professor with uni.student talk **IMP-hit.PST**
‘1. The professor was talking to the student (yesterday).’
2. Professors (usually/generally) talked to students.’
- c. (*emruz/mæmulæn/kol:æn*) *ostad ba danešdzu hæf mi-zæn-e*
today/usually/generally professor with uni.student talk **IMP-hit.NPST-3.SG**
‘1. The professor talks to the student (today).’
2. Professors (usually/generally) talk to students.’

Finally, definiteness in Persian is also affected by the choice of particular lexical items and speaker/listener’s prior expectations regarding the uniqueness or familiarity of their referents. Consider the minimal pairs listed in (12) that compare the effect of *pesær* ‘boy/son’ vs. *pedær* ‘father’ on definite interpretations within the same utterance in (12a), as well as *setare* ‘star’ vs. *xorfid* ‘sun’ in (12b). When the noun *pesær* ‘boy/son’ is used, the more likely interpretation is the indefinite one: there are boys in the alley. This interpretation is typically accompanied by no intonational break between the subject and the verb phrase. It is also possible to have an intonational break after the subject ‘boy/son’ followed by stressing “in alley” to convey a definite interpretation: the boy is in the alley. On the other hand, when we use *pedær* ‘father’, the most likely interpretation is definite: the father is in the alley. The indefinite interpretation is not impossible but requires considerable contextual support. Similarly, when we use the noun *setare* ‘star’, the indefinite interpretation is more prominent and the definite one requires special contextual support. On the other hand with the noun *xorfid* ‘sun’, the only available interpretation is the definite one, unless we are willing to imagine a world with multiple suns. The definiteness bias of

¹The utterance may sound less unacceptable with *kol:æn* “generally” because the adverb allows for a non-generic interpretation too: “The professor talked to the student only.” But the generic interpretation is unacceptable and unavailable.

lexical items like *xorfid* ‘sun’ or *pedær* ‘father’ likely stems from the fact that they often refer to unique and/or familiar referents in the utterance context. As speakers, we have prior expectations regarding the most likely interpretation of such nouns in a new context. While the role of lexical items on definiteness may be clear for words used in (12), prior expectations of definiteness likely exists for almost all nouns and affect our metalinguistic intuitions and judgments.

(12) The effect of lexical items on definiteness:

- a. *pesær* ‘boy/son’ vs. *pedær* ‘father’:
 - i. *pesær tu kutʃ̌(e)-æs(t)*
boy/son in alley-be.3.SG
‘1. There are boys in the alley.
2. The boy is in the alley.’
 - ii. *pedær tu kutʃ̌(e)-æs(t)*
father in alley-be.3.SG
‘1. Father is in the alley.
2. ? There are fathers in the alley.’
- b. *setare* ‘star’ vs. *xorfid* ‘sun’:
 - i. *setare tu asemun-e*
star in sky-be.3.SG
‘1. There are stars in the sky.
2. The star is in the sky.’
 - ii. *xorfid tu asemun-e*
sun in sky-be.3.SG
‘The sun is in the sky.’

3 Definiteness in Plural Nominals

With the complex picture of definiteness in Persian singular nominals in mind, this section presents empirical observations on definiteness in five plural constructions (Table 1). My focus will be mostly on the plural morpheme *-ba* and its semantic contribution. The acceptability or unacceptability of the plural morpheme in the constructions studied here as well as the range of interpretations available for these constructions will constitute the empirical data that I aim to describe using the formal semantic analysis presented in Section 4. Even though the formal account presented in this paper falls significantly short of providing a comprehensive analysis for all these plural constructions, I believe it provides a good starting point.

3.1 Bare Plurals

English bare plurals can have indefinite, habitual, or generic interpretations (Carlson 1977). Persian bare plurals (N-*ba*) can be interpreted as definite in addition to indefinite, habitual, and generic. The exact interpretation depends on many factors including the presence or absence of indefinite morphemes, tense and aspect, phrasal stress, and world knowledge. This is quite similar to what we saw with definiteness in singular nominals in the previous section. The set of examples in this section (13-15)

Table 1: The summary of the six plural constructions studied in this paper

Construction Name	General Schema	Example Noun Phrase
Bare Plurals	N- <i>ha</i>	<i>ketab-a</i> book-PL 'books'
Classifier Plurals	NUM+CL+N- <i>ha</i>	<i>do ta ketab-a</i> two CL book-PL 'the two books'
Indefinite Plurals	(<i>je</i>)+N- <i>ha+i</i>	<i>je ketab-a-i</i> one book-PL-IND 'some books'
Negative Plurals	NEG+N- <i>ha+i</i>	<i>hiŋ ketab(*-a)-i</i> no book-PL-IND 'no book(s)'
Plural Constituent Questions	Wh- <i>ha</i>	<i>ki-a</i> who-PL 'who all?'

show the interaction of tense and aspect with definiteness in the interpretation of bare plurals. Since it is sometimes claimed that syntactic position plays a role in the interpretation of bare plurals, I also control for the syntactic position in these examples.

(13) Subject Position:

- a. (*diruz/#mæmulæn/?kol:æn*) *daneŋdʒu-ha hæf zæd-æn*
yesterday/usually/generally student-PL talk hit.PST-3.PL
'The students talked (yesterday).'
- b. (*diruz/mæmulæn/?kol:æn*) *daneŋdʒu-ha hæf mi-zæd-æn*
yesterday/usually/generally student-PL talk IMP-hit.PST-3.PL
'1. The students were talking (yesterday).
2. Students (usually/?generally) were talking.'
- c. (*emruz/mæmulæn/kol:æn*) *daneŋdʒu-ha hæf mi-zæn-æn*
today/usually/generally student-PL talk IMP-hit.NPST-3.PL
'1. Students (usually/generally) talk.
2. The students talk (today).'

The examples in (13) show the interpretation of a bare plural noun in the subject position of a past perfective sentence (13a), a past imperfective sentence (13b), and a non-past imperfective sentence (13c). In (13a) when the verb stem is in the past tense and does not bear the imperfective prefix, the subject nominal receives a definite interpretation: "the students talked (yesterday)". We can optionally use the adverbial *diruz* "yesterday" to access a past episodic interpretation. However, using the adverbials *mæmulæn* "usually" and *kol:æn* "generally" make the sentence unacceptable. Now compare this to example (13b) where the verb is minimally changed to bear the imperfective prefix *mi-*. Almost all three adverbials are now acceptable and bring out the definite, habitual, or (past) generic interpretations. The

adverbial *kol:æn* “generally” may not be fully acceptable, possibly because it is much more natural when generalizing in present tense. In (13c) where the verb is changed to a non-past stem, all three adverbials are acceptable and the one with *kol:æn* “generally” expresses the generalization that: talking is what students do.

(14) Subject and Indirect Object Positions:

- a. (*diruz/#mæmulæn/?kol:æn*) *ostad-a ba danesd̥ʒu-ha hæf zæd-æn*
yesterday/usually/generally professor-PL with student-PL talk hit.PST-3.PL
‘The professors talked to the students (yesterday).’
- b. (*diruz/mæmulæn/?kol:æn*) *ostad-a ba danesd̥ʒu-ha hæf mi-zæd-æn*
yesterday/usually/generally professor-PL with student-PL talk IMP-hit.PST-3.PL
‘1. The professors were talking to the students (yesterday).
2. Professors were (usually/?generally) talking to students.’
- c. (*emruz/mæmulæn/kol:æn*) *ostad-a ba danesd̥ʒu-ha hæf mi-zæn-æn*
today/usually/generally professor-PL with student-PL talk IMP-hit.NPST-1.SG
‘1. The professors talk to the students (today).
2. Professors (usually/generally) talk to students.’

The examples in (14) show the same pattern of interpretations as we saw in (13), but for the subject and indirect object positions. The presence of the past tense verb stem and the absence of the imperfective morpheme in (22a) seem to generate definite interpretations for both the subject bare plural and the indirect object bare plural. It is unacceptable to use a temporal adverbial like *mæmulæn* “usually” or *kol:æn* “generally” in this sentence. As in the previous examples, it is possible to use *kol:æn* “generally” to mean something different, namely “The professors only talked to the students”, but this is not the generic interpretation intended for our purposes. In (22b), the presence of the imperfective morpheme allows for the habitual interpretation and possibly a past generic interpretation like “Professors used to talk to students generally”. With both the imperfective aspect morpheme and the non-past stem of the verb in (22c), we can easily get all three interpretations (definite, habitual, and generic) for both the subject bare plural and the indirect object bare plural.

(15) Subject and Direct Object Positions:

- a. † *ostad-a danesd̥ʒu-ha be hæf ovord-æn*
professor-PL student-PL to talk bring.PST-3.PL
‘The professors made so many students talk.’
- b. (*diruz/#mæmulæn/?kol:æn*) *ostad-a danesd̥ʒu-ha ro be hæf ovord-æn*
yesterday/usually/generally professor-PL student-PL ACC to talk bring.PST-3.PL
‘The professors made the students talk (yesterday).’
- c. (*?diruz/mæmulæn/?kol:æn*) *ostad-a danesd̥ʒu-ha ro be hæf mi-ovord-æn*
yesterday/usually/generally professor-PL student-PL ACC to talk IMP-bring.PST-3.PL
‘1. The professors were making the students talk (yesterday).
2. Professors (usually/generally) were making the students talk.’
- d. (*emruz/mæmulæn/kol:æn*) *ostad-a danesd̥ʒu-ha ro be hæf mi-ar-æn*
yesterday/usually/generally professor-PL student-PL ACC to talk IMP-bring.PST-3.PL

- ‘1. The professors make the students talk (today).
2. Professors (usually/generally) make students talk.’

Examples in (15) show that the same pattern of judgments holds for bare plurals in the subject and direct object positions. First, in the direct object position, we have the option of not using the object marker (15a) or using the object marker (15b). Without the object marker, the sentence can only be interpreted as “The professors made a lot of students talk” if accompanied by a special intonation. This construction and interpretation is rather archaic and arguably not an active part of modern colloquial Persian so I have marked it with a dagger symbol. Setting this issue aside, examples (15b) to (15d) show the same pattern we saw before in the presence of the object marker: without the imperfective morpheme in (15b), the bare plurals are interpreted as definite and with the imperfective morpheme in (15c) and (15d), the habitual and/or the generic interpretations become available as well.

I should note that while the example in (15a) sounds archaic and unnatural, there are many other examples of bare plurals with the similar indefinite interpretations of “many” or “a lot” that sound natural and are used in colloquial Persian. The examples in (16) are shorter forms of sentences I have observed in social media posts.

- (16) a. *sal-ha gozæfte væli hænuz yad-ej-e*
 year-PL pass.PST.3.SG but still memory-3.SG-be.3.SG
 ‘Years have passed but he still remembers.’
- b. *emfæb bæde mah-ha ræft-æm mæGæze-gærdi*
 tonight after year-PL go.PST-1.SG shopping
 ‘After months, I went shopping tonight.’
- c. *Ali sæxti-ha kefid ta doktor fod*
 Ali hardship-PL pull until doctor become.3.SG
 ‘Ali had so many difficulties until he became a doctor.’

Mirrazi (2021) investigates Persian bare plurals and makes the important observation that modification can improve the acceptability of certain bare plurals. She presents a wide range of examples like (17) that show the acceptability of bare nominals (17a), followed by the unacceptability of bare plurals in the same sentence (17b), and the improved acceptability of the modified bare plural in the same sentence (17c). She argues that this pattern can be explained by assuming that the plural marker makes the bare nominal definite, while modification adds a situation variable which allows for the existential reading of a definite.

- (17) a. *ma daneſďzu hæst-im*
 1.PL uni.student be.NPST-1.PL
 ‘We are students.’
- b. * *ma daneſďzu-ha hæst-im*
 1.PL uni.student-PL be.NPST-1.PL
 ‘We are students.’
- c. *ma daneſďzu-ha-je zæbanſenasi hæst-im*
 1.PL uni.student-PL-EZ linguistics be.NPST-1.PL
 ‘We are linguistics students.’

(Mirrazi 2021)

In Jasbi (2015), I had provided a similar analysis to Mirrazi (2021) in which the plural morpheme made nouns definite and was a MAX operator. But as we shall see in the next section, this analysis is incompatible with a large body of empirical data in which the plural morpheme appears on nouns marked by indefinite morphemes. It is also important to note that not any type of modification seems to be fine for the existential interpretation of a bare plural. The data in (18) below shows that while the plural noun “students” is acceptable when modified by “undergraduate” in (18a), it is not as acceptable if modified by “good” or “studious” in (18b) and (18c). The noun phrases in (18b) and (18c) need the indefinite clitic *-i* while the noun phrase in (18a) is unacceptable when followed by the indefinite clitic. My guess is that modification helps the bare plural if it creates a recognizable sub-kind or sub-type (e.g. linguistics students, undergraduate students) but I leave a proper treatment of modification in bare plurals to future research.

- (18) a. *ma danefdʒu-ba-je lisans hæst-im*
 1.PL uni.student-PL-EZ undergraduate be.NPST-1.PL
 ‘We are undergraduate students.’
 b. * *ma danefdʒu-ba-je xub hæst-im*
 1.PL uni.student-PL-EZ good be.NPST-1.PL
 ‘We are good students.’
 c. * *ma danefdʒu-ba-je daærsxun hæst-im*
 1.PL uni.student-PL-EZ studious be.NPST-1.PL
 ‘We are studious students.’

3.2 Classifier Plurals

Singular indefinites in Persian are often expressed using the numeral *je(k)* ‘one’ as example (19a) shows. In some varieties of Persian such as Yazdi *je(k)* ‘one’ can be followed by the classifier *ta*, while in Tehrani or other varieties using this classifier with *je(k)* is unacceptable. Since the use of classifier *ta* after ‘one’ is subject to dielactal variation in Persian I have used the % symbol to mark its (un)grammaticality. Persian can also pluralize nouns by using numerals like , *do* ‘two’, *se* ‘three’ shown in (19b), or the “unknown numeral” *iʃænd* in (19c), followed by a classifier like *ta*, without using the plural suffix. It is sometimes said that the classifier *ta* in such examples is “optional”. This does not seem entirely accurate to me. In colloquial Persian the use of the classifier *ta* (or a more specific classifier) in examples like (19b) and (19c) seems obligatory. Dropping the classifier in these examples would be unacceptable for everyday speech, but it is acceptable in the more formal and archaic registers of Persian.

- (19) Examples of the numeral+classifier construction:
 a. *je (%ta) danefdʒu hæfzæd*
 one (CL) uni.student talk hit.PST
 ‘A student talked.’
 b. *do/se/iʃar/pændʒ/... ta danefdʒu hæfzæd-æn*
 two/three/four/five/... CL uni.student talk hit.PST-3.PL
 ‘Two/three/four/five/... students talked.’
 c. *iʃænd ta danefdʒu hæfzæd-æn*
 NUM CL uni.student talk hit.PST-3.PL

‘Some students talked.’

The interpretation of the morpheme *tʃænd* in declarative vs. interrogative contexts is considerably different. In a declarative sentence like (19c), it is interpreted similar to “some” in English, while the same sentence with an interrogative (rising) intonation in (20) receives an interpretation similar to “how many”. Any formal analysis of this morpheme should be able to accommodate the shift in interpretation between its declarative and interrogative forms.

(20) Example of the unknown numeral in a question:

tʃænd ta danɛʃdʒu hæf zæd-æn?
 NUM CL uni.student talk hit.PST-3.PL

‘How many students talked?’

Similar to the previous section, I investigate the interpretation of classifier plurals in different syntactic positions and with different verbal tense and aspect. The examples in (21) are identical to what we had in (13) with bare plurals, except now I use classifier plurals. In these examples I use the numeral *do* “two”, the numeral *se* “three”, and the unknown numeral *tʃænd*. The verb in example (21a) is in the past tense and uses the perfective aspect. The sentence has a plural indefinite interpretation depending on the number used, but it cannot be interpreted as habitual (two/three/some students usually talked) or generic (it’s generally the case that some students talked). When we change the aspect to imperfective in (21b) and then change the tense to non-past in (21c), the habitual interpretation becomes available. It is important to note that the sentences are unacceptable with the adverbial *kol:æn* “generally” if we mean to communicate a generic interpretation. The adverb *kol:æn* can also communicate other interpretations that would make the sentence acceptable such as “all that happened was that some professors talked to some students.” The generic interpretation seems to be only available in (21c) where the verb is imperfective and non-past: “it is generally the case that when you have two/three/some-number-of students, they talk”.

(21) Example classifier plurals in the subject position interacting with tense and aspect:

- a. (*diruz/#mæmulæn/#kol:æn*) *do/se/tʃænd ta danɛʃdʒu hæf zæd-æn*
 yesterday/usually/generally two/three/NUM CL student talk hit.PST-3.PL
 ‘Two/three/some students talked (yesterday).’
- b. (*diruz/mæmulæn/#kol:æn*) *do/se/tʃænd ta danɛʃdʒu hæf mi-zæd-æn*
 yesterday/usually/generally two/three/NUM CL student talk IMP-hit.PST-3.PL
 ‘Two/three/some students talked (yesterday/usually).’
- c. (*emruz/mæmulæn/kol:æn*) *do/se/tʃænd ta danɛʃdʒu hæf mi-zæn-æn*
 today/usually/generally two/three/NUM CL student talk IMP-hit.NPST-3.PL
 ‘Two/three/some students talked (yesterday/usually/generally).’

The examples in (22) test the interpretation of classifier plurals in subject and indirect object positions. I have dropped exact numerals like “two” and “three” in this example and only included the unknown numeral for the sake of convenience. The judgments do not change if we use the exact numerals as far as I can tell. The pattern of interpretations is similar to what we saw in (21). The habitual and generic interpretations are not available when the verb is past perfective while the habitual interpretation is available when we use the imperfective morpheme on the verb. The generic interpretation

is mainly available in non-past imperfective sentence: “It is generally the case that when you have some professors and some students, what they do is talk”.

- (22) Example classifier plurals in subject and indirect object positions interacting with tense and aspect:

a. (*diruz/#mæmulæn/#kol:æn*) *tʃænd ta ostad ba tʃænd ta danɛfɔ̌zu hærf*
 yesterday/usually/generally NUM CL professor with NUM CL student talk
zæd-æn
 hit.PST-3.PL

‘Some professors talked to some students (yesterday).’

b. (*diruz/mæmulæn/#kol:æn*) *tʃænd ta ostad ba tʃænd ta danɛfɔ̌zu hærf*
 yesterday/usually/generally NUM CL professor with NUM CL student talk
mi-zæd-æn
 IMP-hit.PST-3.PL

‘Some professors talked to some students (yesterday/usually).’

c. (*emruz/mæmulæn/kol:æn*) *tʃænd ta ostad ba tʃænd ta danɛfɔ̌zu hærf*
 today/usually/generally NUM CL professor with NUM CL student talk
mi-zæn-æn
 IMP-hit.NPST-I.SG

‘Some professors talked to some students (yesterday/usually/generally).’

In (23) I test the interpretation of classifier plurals in the subject and direct object positions. Again in a similar pattern to the examples above we see that in (23a) when the verb is past perfective, the classifier plurals receive indefinite interpretations and the habitual and generic readings are unacceptable. If we add the imperfective morpheme to the verb as (23b) shows, the habitual interpretation is also possible: “Some professors usually made some students talk”. Finally when we change the verb stem to be non-past, the generic interpretation is also possible: “It is generally the case that some professors make some students talk.” The object marker is obligatory on all these examples but this is the property of the chosen verb for this example. If we change the verb to something like “see”, or “brought”, it is possible to drop the object marker.

- (23) Example classifier plurals in subject and direct object positions interacting with tense and aspect:

a. (*diruz/#mæmulæn/#kol:æn*) *tʃænd ta ostad tʃænd ta danɛfɔ̌zu ro be hærf*
 yesterday/usually/generally NUM CL professor NUM CL student ACC to talk
ovord-æn
 bring.PST-3.PL

‘Some professors made some students talk (yesterday).’

b. (*diruz/mæmulæn/#kol:æn*) *tʃænd ta ostad tʃænd ta danɛfɔ̌zu ro be hærf*
 yesterday/usually/generally NUM CL professor NUM CL student ACC to talk
mi-ovord-æn
 IMP-bring.PST-3.PL

‘Some professors made some students talk (yesterday/usually).’

- c. (*emruz/mæmulæn/kol:æn*) *tʃænd ta ostad tʃænd ta daneʃdʒu ro be hæf*
 yesterday/usually/generally NUM CL professor NUM CL student ACC to talk
mi-ar-æn
 IMP-bring.PST-3.PL
 ‘Some professors make some students talk (today/usually/generally).’

Comparing the examples above to their counterparts with bare plurals, we can conclude that classifier plurals in Persian do not lend themselves easily to definite interpretations. It is tempting to say that classifier plurals are indefinite but there are systematic counterexamples to this generalization. As (24) shows, the demonstrative determiners can modify classifier plurals and result in a definite interpretation.

(24) Examples of numerals with demonstratives:

- a. *in/un je daneʃdʒu hæf zæd*
 this/that one student-PL talk hit.PST
 ‘This/that one student talked.’
- b. *in/un do/se/.../tʃænd ta daneʃdʒu hæf zæd-æn*
 this/that two/three/... CL student talk hit.PST-3.PL
 ‘These/those two/three/.../some students talked.’

While classifier plurals can convey plurality on their own, they can also combine with the plural suffix *-ba*. Of course, the numeral *je(k)* ‘one’ is completely unacceptable with the plural suffix as (25a) shows. However, other numerals like *do* ‘two’ or *se* ‘three’ shown in example (25b) are acceptable with the plural morpheme, although the plural morpheme may sound a bit redundant and be more felicitous when there is a communicative need for emphasis on plurality (as commented by some of my informants). The combination of the unknown numeral and the plural suffix in (25c) seems to be less acceptable than the exact numerals in (25b). The demonstratives are optional and seem to help the acceptability of the sentences when present. Finally, the examples in (26) are selected from natural interactions online to show cases where the numeral+classifier construction and the plural suffix are used together.

(25) Examples of numerals with the plural suffix:

- a. * (*in/un*) *je daneʃdʒu-ba hæf zæd*
 this/that one student-PL talk hit.PST
 ‘* (this/that) one students talked.’
- b. (*in/un*) *do/se/... ta daneʃdʒu-ba hæf zæd-æn*
 (this/that two/three/... CL student talk hit.PST-3.PL
 ‘These/those/the two/three/... students talked.’
- c. ? (*in/un*) *tʃænd ta daneʃdʒu-ba hæf zæd-æn*
 this/that NUM CL student-PL talk hit.PST-3.PL
 ‘This/that/the some-number-of students talked.’

(26) Corpus examples of numerals with the plural suffix:

- a. *hamin tʃænd ta podcast-a o ketab-a hæm basævad-et næ-kærd-ø*
 these NUM CL podcast-PL and book-PL too learned-2.SG NEG-do.PST-3.PL
 ‘Not even these podcats or books made you knowledgeable.’
- b. *fife je tʃænd ta maʃin-a fɪkæst-e*
 glass EZ NUM CL car-PL break.PST-PERF.3.SG
 ‘Some cars have broken windows.’
- c. *tʃænd ta kar-a je æGæboftad(e)-æm o ændʒam dad-æm*
 NUM CL work-PL EZ late-1.SG ACC finish give.PST-1.SG
 ‘I finished some of my overdue tasks.’

3.3 Indefinite Plurals

The indefinite clitic *-i* can appear on plurals with *-ba* to form what I call an indefinite plural. Similar to bare nouns modified by the indefinite clitic, there are restrictions on the distribution of plurals with *-i*. The clitic cannot make a plural indefinite on its own. The examples in (27) are unacceptable without intonational or contextual support in colloquial Persian. There is a way to rescue them by reading them with a special intonation that is typically used if the indefinite plural is preceded by *tʃe* “what”, with the meaning “what weird/strange sounds!” or “what good/bad friends, I have!”. But these interpretations are not the same as simple indefinites anymore.

(27) Indefinite plural examples in subject, direct object, and PP object positions:

- a. # *seda-ba-i mi-a-d*
 sound-PL-IND IMP-come.NPST-3.SG
 ‘There are some strange sounds.’
- b. # *tʃiz-a-i xærid-æm*
 thing-PL-IND buy.PST-1.SG
 ‘I bought some things.’
- c. # *Mæɾjæm æz heivuna-i mi-tærs-e*
 Maryam from animal-IND IMP-fear.NPST-3.SG
 ‘Maryam is afraid of some animals.’

Similar to what we saw in bare nouns with the clitic, the indefinite plurals in (27) become acceptable if they are further modified by an adjective as in (28) or by relative clauses as in (29). It is important to note that a signature behavior of *any* in English is also the fact that modification improves its acceptability. This phenomenon is sometimes referred to as “licensing by modification” or “subtriggering” (Dayal 1988, 2004; LeGrand 1975).

(28) Indefinite plural examples with adjectival modification:

- a. *seda-ba-je ædʒib-i mi-a-d*
 sound-PL-EZ strange-IND IMP-come.NPST-3.SG
 ‘There are some strange sounds.’
- b. *tʃiz-a-je xub-i xærid-æm*
 thing-PL-EZ good-IND buy.PST-1.SG
 ‘I bought some good things.’

- c. *Mærjæm æz heivun-a-je xas-i mi-tærs-e*
 Maryam from animal-PL-EZ specific-IND IMP-fear.NPST-3.SG
 ‘Maryam is afraid of some specific animals.’

(29) Indefinite plural examples with relative clause modification:

- a. *seda-ba-i mi-a-d ke ta bala næ-fnid-æm*
 sound-PL-IND IMP-come.NPST-3.SG that until now NEG-hear.PST-1.SG
 ‘I hear sounds I have not heard before.’
- b. *t̪ɪz-a-je xærid-æm ke hɪf-ki næ-dar-e*
 thing-PL-EZ buy.PST-1.SG that no-one NEG-have.NPST-3.SG
 ‘I bought some things that no-one has.’
- c. *Mærjæm æz heivun-a-i mi-tærs-e ke parvaz mi-kon-æn*
 Maryam from animal-PL-IND IMP-fear.NPST-3.SG that fly IMP-do.NPST-3.PL
 ‘Maryam is afraid of animals that fly.’

It is possible to add the numeral *je(k)* ‘one’ to make indefinite plurals acceptable. While the examples in (27) were unacceptable, their minimal variants with *je(k)* ‘one’ in (30) are completely acceptable. This is rather surprising. Under normal assumptions regarding plurality and numerals, the combination of a numeral like “one” and a plural noun like “animals” should result in a contradictory and unacceptable form: “one animals”. In fact, this is exactly what we find if the indefinite clitic *-i* is not on the noun as shown in (31). Therefore, it is reasonable to assume that the semantic effect of the indefinite clitic, allows for further combination with the numeral “one” in ways that does not result in a contradiction or unacceptability.

(30) Indefinite plurals combining with numeral *je(k)*:

- a. *je seda-ba-i mi-a-d*
 one sound-PL-IND IMP-come.NPST-3.SG
 ‘There are some sounds.’
- b. *je t̪ɪz-a-i xærid-æm*
 one thing-PL-IND buy.PST-1.SG
 ‘I bought some things.’
- c. *Mærjæm æz je heivun-a-i mi-tærs-e*
 Maryam from one animal-PL-IND IMP-fear.NPST-3.SG
 ‘Maryam is afraid of some animals.’

(31) Bare plurals combining with numeral *je(k)*:

- a. *#je seda-ba mi-a-d*
 one sound-PL IMP-come.NPST-3.SG
 ‘There are some sounds.’
- b. *#je t̪ɪz-a xærid-æm*
 one thing-PL buy.PST-1.SG
 ‘I bought some things.’

- c. # *Mærjæm æz je heivun-a mi-tærs-e*
 Maryam from one animal-PL IMP-fear.NPST-3.SG
 ‘Maryam is afraid of some animals.’

Finally, modifying indefinite plurals by placing them in downward entailing or non-veridical environments (Giannakidou 1998) does not improve their acceptability as the examples in (32) show. This is the opposite of what we see with bare nouns that are marked by the indefinite clitic. Placing bare nouns with *-i* in downward entailing or non-veridical environments makes the utterance acceptable as (33) shows. This observation suggests that the Persian plural morpheme is not fully compatible with downward-entailing or non-veridical environments. We will investigate this matter more in the section on negative and non-veridical environments.

(32) Examples of indefinite plurals in non-veridical environments:

- a. # *seda-ha-i ne-mi-a-d*
 sound-PL-IND IMP-come.NPST-3.SG
 ‘There are no sounds / Some sounds are not heard.’
- b. # *seda-ha-i mi-a-d?*
 sound-PL-IND IMP-come.NPST-3.SG
 ‘Are there some sounds?’
- c. # *æge seda-ha-i mi-a-d, be-gu!*
 if sound-PL-IND IMP-come.NPST-3.SG, SUB-say.NPST
 ‘If there are sounds, tell me/us!’

(33) Examples of bare nominals with the clitic in non-veridical environments:

- a. *seda-i ne-mi-a-d*
 sound-IND IMP-come.NPST-3.SG
 ‘There are no sounds.’
- b. *seda-i mi-a-d?*
 sound-IND IMP-come.NPST-3.SG
 ‘Are there any sounds?’
- c. *æge seda-i mi-a-d, be-gu!*
 if sound-IND IMP-come.NPST-3.SG, SUB-say.NPST
 ‘If there are any sounds/noise, tell him that me/us!’

3.4 Negative Plurals and Plurals in Non-Veridical Environments

The interpretation of plurals in downward-entailing or non-veridical environments has been very influential in shaping the formal semantics of plurality (Farkas & de Swart 2010; Giannakidou 1998; Spector 2007). English plurals can receive an exclusive or inclusive interpretation depending on the linguistic environment. The examples in (34) repeated and adapted from Farkas & de Swart (2010) demonstrate this point. The plural examples in (34a) are true if more than one horse was seen, and false if only one horse or no horse was seen. We can say that their interpretation excludes singular cases where a single horse was seen. We may naturally expect that the negation of such plurals would be true when no horse or only one horse was seen and false otherwise. However, examples in (34b) show that this is not the

case for negation and other logical operations. In all these examples, a possible and often dominant interpretation is one in which the plural noun ‘horses’ includes singular cases too and for example in the case of negation, denies any horses, singular or plural, having been seen. A lot of research in formal semantics has focused on explaining this behavior of plurals in negative or non-veridical environments.

- (34) a. Plural Exclusive Interpretation:
- i. Some horses have been seen in this meadow.
 - ii. Mary has seen horses in this meadow.
 - iii. Sam has always seen horses in this meadow.
- b. Plural Inclusive Interpretation:
- i. No horses have ever been seen in this meadow.
 - ii. Have you ever seen horses in this meadow?
 - iii. If you have ever seen horses in this meadow, you should call us.
 - iv. Sam has never seen horses in this meadow.

Perhaps the most important observation when it comes to similar Persian constructions is that the most natural Persian translation of the English sentences in (34) does not involve plurals at all but rather bare nouns as shown in (35). However, the meanings of the bare nouns in (35) are slightly different from the English plurals in that the Persian sentences are true whether a single horse or multiple horses were seen. The bare noun seems to be truly number neutral (Ghomeshi 2008). This does not mean we cannot use plural indefinites for such sentences. The examples in (36) show that with the right type of support from the numeral *je(k)* and the indefinite clitic, or sometimes modification with adjectives, it is possible to use the plural suffix on the noun ‘horse’. However, these examples have additional implications and are further away from the original English examples. For example, the sentence in (36a) implies that the speaker does not know the type of horses exactly or does not want to share it.

Most importantly, when embedded in non-veridical or downward entailing environments, the use of the plural suffix becomes unacceptable. Consider the examples listed in (37). As far as I can tell they are all acceptable without the plural suffix and unacceptable when the plural morpheme is used. More generally, the plural suffix is completely unacceptable with a negative quantifier as (35a) shows unlike what we find in English negative plurals like “no horses” or “no books”. This further confirms the observation in the previous section with example (32) that showed the plural morpheme is not acceptable in non-veridical or downward entailing environments.

- (35) a. *æsb tu in tʃæmænzar dide fod-e*
horse in this meadow seen become-3.SG
‘Horses have been seen in this meadow.’
- b. *Mærjem tu in tʃæmænzar æsb did-e*
Maryam in this meadow horse see.PST-3.SG
‘Maryam has seen horses in this meadow.’
- c. *tʃænd ta æsb tu in tʃæmænzar dide fod-e*
NUM CL horse in this meadow seen become-3.SG
‘Some horses have been seen in this meadow.’

- d. *Mærjæm hæmife tu in tʃæmænzar æsb mi-bin-e*
 Maryam always in this meadow horse IMP-see.PST-3.SG
 ‘Maryam always sees horses in this meadow.’
- (36) a. *je æsb-a-i tu in tʃæmænzar dide fod-æn*
 one horse-PL-IND in this meadow seen become-3.PL
 ‘Some (or other) horses have been seen in this meadow.’
- b. *Mærjæm tu in tʃæmænzar æsb-a je ziad-i did-e*
 Maryam in this meadow horse-PL EZ alot-IND see.PST-3.SG
 ‘Maryam has seen many horses in this meadow.’
- c. *Mærjæm hæmife tu in tʃæmænzar æsb-a je Gæfæng-i mi-bin-e*
 Maryam always in this meadow horse-PL EZ beautiful-IND IMP-see.PST-3.SG
 ‘Maryam always sees beautiful horses in this meadow.’
- (37) Plural suffix in non-veridical and downward entailing environments:
- a. *Mærjæm tu in tʃæmænzar hitʃ æsb(*-ha)-i næ-did-e*
 Maryam in this meadow nothing horse-PL-IND NEG-see.PST-3.SG
 ‘Maryam has seen no horses in the meadow.’
- b. *ta hala tu in tʃæmænzar æsb(*-ha)-i did-i?*
 till now in this meadow horse-PL-IND see.PST-3.SG
 ‘Have you ever seen a horse in this meadow?’
- c. *age ta hala tu in tʃæmænzar æsb(*-ha) did-i, be ma be-gu!*
 if till now in this meadow horse-PL-IND see.PST-3.SG, to 2.PL SUB-say.NPST!
 ‘If you have ever seen horses in this meadow, tell us!’
- d. *hær ki tu in tʃæmænzar æsb(*-ha)-i did-e, be ma be-ge!*
 all who in this meadow horse-PL-IND see.PST-3.SG, to 2.PL SUB-say.NPST!
 ‘Everyone who has seen horses in this meadow, should tell us!’

The main takeaway is that Persian bare nouns often do what English bare plurals do. Especially in non-veridical and downward entailing environments, bare nouns are more natural in Persian than plurals with *(h)a*. Therefore, it is possible that while in English the plural morpheme can have both an inclusive and exclusive plural interpretation, in Persian the plural morpheme refers to plural entities and only has an exclusive interpretation. This is likely because bare nouns can and often do refer to both singular and plural entities; an option not available in English.

3.5 Constituent Question Plurals

The plural suffix can also combine with constituent question words such as *ki* ‘who’ in (38a), *tʃi* ‘what’ in (38b), *kej* ‘when’ in (38c), *kodʒa* ‘where’ (38d), *kodum* ‘which’ (38e), but not *tʃera* ‘why’ and *tʃetor* ‘how’ shown with an asterisk in (38f) and (38g). Therefore with Persian question words equivalent to *who*, *what*, *when*, *where*, and *which*, both unmarked and plural marked forms are acceptable but for *why* and *how* only the unmarked form is acceptable. The plural forms of the question words imply plurality but not definiteness. In other words, the plural form of ‘what’, ‘who’, ‘when’, or ‘where’ is not equivalent in

meaning to ‘which of the things’, ‘which of the people’, ‘which of the times’, or ‘which of the places’. There is no implication that the answer is coming from a mutually known set of items.

(38) Examples for plural marking of question words in Persian:

- a. *ki(-a) ro did-i?*
who(-PL) ACC see.PST-2.SG
‘Who did you see?’
- b. *tī(-a) xærid-i?*
what(-PL) buy.PST-2.SG
‘What did you buy?’
- c. *kej(-a) mi-r-i baʃgah?*
when(-PL) IMP-go.NPST-2.SG gym
‘When do you go to the gym?’
- d. *kodʒa(-ha) mi-r-i?*
where(-PL) IMP-go.NPST-2.SG
‘Where are you going?’
- e. *kodum(-a) ro mi-bæ-r-i?*
which(-PL) ACC IMP-take.NPST-2.SG
‘Which one(s) do you take?’
- f. *tjæra(*-ha) ræft-i?*
why(-PL) go.PST-2.SG
‘Why did you go?’
- g. *tjætori(*-ha) ræft-i?*
why(-PL) go.PST-2.SG
‘How did you go?’

However, the plural variants of these question words do seem to imply that the answer is plural, or at least the speaker expects the answer to be plural. To test the source of this implication as well as its strength, I select contexts that require uniqueness of the answer and contexts that allow for singular or plural answers. I also test the same question word with or without the plural suffix in both contexts. If the plural suffix strongly implies plurality, it should be unacceptable in contexts where a unique answer is expected, but acceptable in contexts where there is no such expectation.

Examples in (39) show the behavior of the question word *kodʒa* ‘where’ in the context of asking about the location of a well-known mountain which requires a unique answer (39a), vs. the location of a horse or multiple horses, a context with possible singular or plural answers (39b). In each context, the two sentences are minimal pairs such that the question word ‘where’ in one sentence is bare (39a-i, 39b-i) and the question word in the other sentence takes the plural suffix (39a-ii, 39b-ii). As the judgments show when there is a unique entity under discussion (Mt. Damavand), the plural morpheme on the question word is unacceptable, because the same mountain cannot be in multiple places. But when the entity under discussion can be any number (e.g. horses), the question word may take the plural suffix *-ha* and imply that there are multiple locations for the entities under discussion.

(39) The interaction of answer uniqueness with plural marking of the question word:

- a. Context: necessary unique answer
 - i. *Golle je dæmavænd kodʒa-st?*
summit EZ Damavand where-3.SG
'Where is Mt. Damavand?'
 - ii. # *Golle je dæmavænd kodʒa-ha-st?*
summit EZ Damavand where-PL-3.SG
'Where are Mt. Damavand?'
- b. Context: possible singular or plural answer
 - i. *æsb kodʒa-st?*
horse where-3.SG
'1. Where is the horse? 2. Where are horses (generally)?'
 - ii. *æsb kodʒa-ha-st?*
horse where-PL-3.SG
'Where are horses (generally)?'

Similarly in example (40) below I test the question word *kej* 'when' and its plural form in a context that requires a unique answer (40a), namely someone's birthday as well as a context that can have a singular or plural answer, namely going to bed (40b). The plural form again is unacceptable when the context requires uniqueness of the answer while the bare form with no plural morpheme is acceptable in both. These observations suggest to me that while definiteness is not an implication of the plural morpheme, plurality is indeed its implication even in the question environment.

(40) The interaction of answer uniqueness with plural marking of the question word:

- a. Context: necessary unique answer
 - i. *kej be donja umæd-i?*
when to world come.PST-2.SG
'When were you born?'
 - ii. # *kej-a be donja umæd-i?*
when-PL to world come.PST-2.SG
'What times were you born?'
- b. Context: possible singular or plural answer
 - i. *kej mi-xab-i?*
when IMP-sleep-2.SG
'When do you sleep? (when are you going to sleep?)'
 - ii. *kej-a mi-xab-i?*
when-PL IMP-sleep-2.SG
'What times do you (usually) sleep?'

4 A Formal Semantic Analysis of Plurality and Definiteness in Persian

In this section I translate some sentences of Tehrani Persian that exemplify the common singular/plural and definite/indefinite constructions into a typed-logical formal language (Carpenter 1998; Coppock

& Champollion 2019; Dowty et al. 1981; Montague 1973). My main goals are first to propose lexical meanings for individual morphemes like the plural marker; second, to represent the compositional structure of these constructions; and third, to clarify the relationship between plurality and definiteness in Persian. The analysis will not cover all the data discussed in the previous section but it has the potential to be extended to at least some of them.

I create a toy model M with the domain D and the interpretation function I : $M = \langle D, I \rangle$. The domain D of our toy model contains some apples and one table. I define three predicates: *APPLE*, *TABLE*, and *ON*. The predicate *APPLE* refers to the set of all apples, their mereological sums, and apple parts like different ways of slicing the apples (Link 1983). This allows for *APPLE* to have singular, plural, and mass predication. This is a departure from prior literature on plurality such as Link (1983) but I will show that it is justified by the facts of Persian. The predicate *ON* is defined as a set of binary relations between entities in the domain D . Finally, I assume that definiteness presupposes existence and uniqueness of the entity described by the nominal (Frege 1952; Strawson 1950), and that failure of these presuppositions results in undefinedness. I use the iota operator (Partee 1986) to formally implement the semantic effects of definiteness.

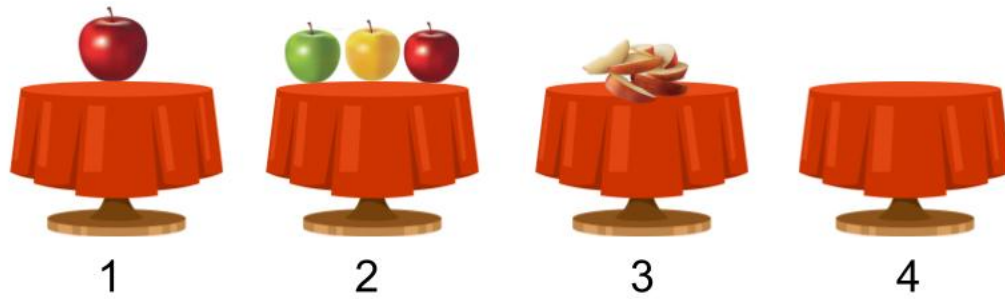


Figure 1: Four scenarios for assessing the truth conditions of the examples in this section. In Scenario 1, there is only one apple on the table. In Scenario 2, three apples are on the table. In Scenario 3 there are apple slices on the table, and in Scenario 4 there is nothing on the table.

I will also use the four scenarios depicted in Figure 1 to talk about the truth conditions of the examples I consider in this section. I start with the bare noun in the subject position². In example (41), the question under discussion (QUD) is “what is on the table”. The reply has a special intonational contour and is true in scenarios 1-3 of Figure 1 but not 4. In other words, example (41) is true whether there is a single apple, multiple apples, or apple parts on the table.

(41) QUD: What is on the table? [ʔi ru miz e?]

[sib]_f ru miz e
apple on table be.3.SG

‘There is apple on the table’

Figure 2 shows the semantic derivation for example (41). The Persian words *sib*, *ru*, and *miz* are translated as the predicates *APPLE*, *ON*, and *TABLE* in our formal language respectively. The agreement

²I discuss the subject position to avoid complexities related to pseudo-incorporation of objects.

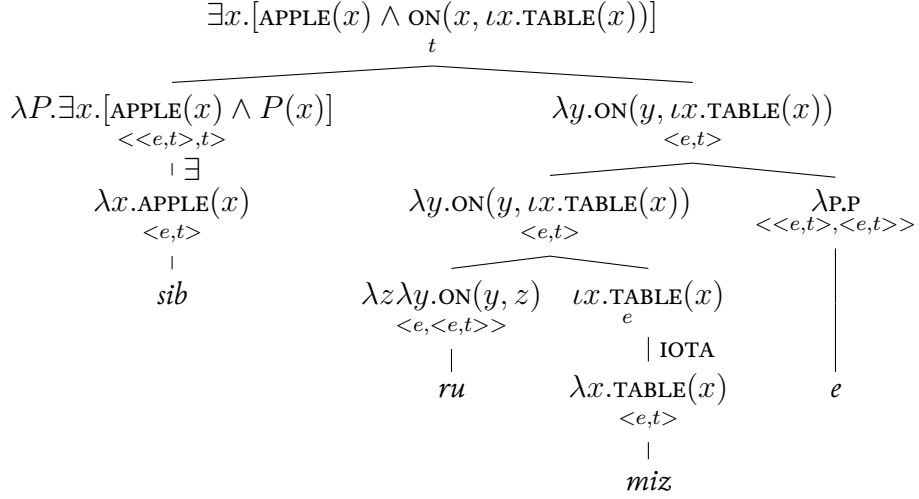


Figure 2: Example derivation tree for the bare nominal construction

morpheme e is translated as an identity function. Similar to the “to be” copula in English, the agreement morpheme e on the verb can convey tense, aspect, and person information but I do not aim to model these phenomena in current work. The tree composition will proceed as expected except for the predicate *APPLE* which undergoes covert type shifting to introduce an existential quantifier. This quantifier then can quantify over singular, plural, and part entities, delivering the intended reading of (41): the statement is true as long as there is some entity, singular, plural, or (apple) part, that is in the relationship denoted by *ON* with the unique table in our model.

Next in (42), I use the same bare nominal construction in (41) but change the QUD which in turn changes the intonational contour and adds stress to the VP “is on the table”. The result is a singular definite, with similar interpretive signature of “the apple” in English when tested against the scenarios of Figure 1: it is true and felicitous in Scenario 1; infelicitous in Scenario 2 (unless the speaker can select a unique apple they intend to refer to); infelicitous in Scenario 3 (unless maybe the apple slices belong to a unique apple the speaker and addressee knew about); and infelicitous in Scenario 4 and possibly false because there are no apples on the table regardless of which one the speaker intends to refer to. These judgments are quite similar to what we expect from definite descriptions.

(42) QUD: Where is the apple? [sib kodʒa-stʔ]

sib [ru miz]_f *e*
apple on table be.3.SG

‘The apple is on the table’

Figure 3 shows the semantic derivation that corresponds to the singular definite construction in (42). The verb phrase composes the same way as before. The main difference is the treatment of the predicate *APPLE*. Under the right set of linguistic and contextual circumstances, including the absence of indefinite morphemes, the presence of appropriate intonation, and background knowledge that biases the uniqueness of the nominal denotation, the nominal can be covertly type-shifted using the iota operator (Partee 1986). I leave a more thorough investigation of factors contributing to definiteness in Persian for future work.

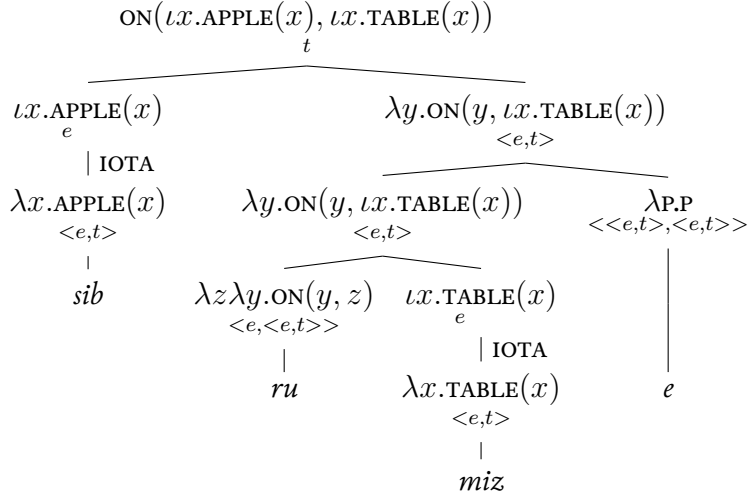


Figure 3: Example derivation tree for the singular definite construction

In (43) I add the plural marker to the bare nominal construction in (42) and change the QUD to ask about plural apples too. The result is similar to the English definite plural construction. It is infelicitous or false in Scenario 1 because there is only one apple on the table, true and felicitous in Scenario 2, infelicitous in Scenario 3 unless we assume the speaker sliced the apples but is still referring to their previous state, and infelicitous or false in Scenario 4 given that there are no apples on the table regardless of which one the speaker intends to refer to.

(43) QUD: Where are the apples? [sib-a kodʒan?]

sib-a [ru miz]_f æn
apple-PL on table be.3.PL

‘The apples are on the table’

Figure 4 shows the semantic derivation for definite plurals and more specifically example (43). The derivation is similar to what we had for singular definites, except that now we have the plural morpheme *ba*. I analyze the plural morpheme as a function from a predicate P to its modified version [⊗]P. I borrow the circled star operation from Link (1983) who proposes it as the pluralization operation that removes atomic entities from the plural set. Unlike Link (1983) I do not take basic predicates like APPLE to denote atomic entities. In my analysis I include pluralities and portions of matter as part of the denotation of the basic predicate. As a result, the circled star operation simply selects the set of plural entities out of the basic predicate denotation, rather than building them out of atomic entities denoted by the singular predicate. Under this analysis, plurals in Persian do not include the atoms unlike the common treatment of English plurals; but also see Chierchia (1998) for a similar proposal for English plurals. In the previous sections I have presented several observations that suggested Persian plurals do not refer to atoms in non-veridical environments unlike English plurals.

The analysis above has the major advantage that it can accommodate both definite and indefinite interpretations of bare plurals in Persian. However, it has the disadvantage that it is too flexible. For example in Figure 4, nothing formally stops us to covertly type-shift the plural noun to derive an indefinite plural. However, our intuitions suggest an indefinite reading is not available for example

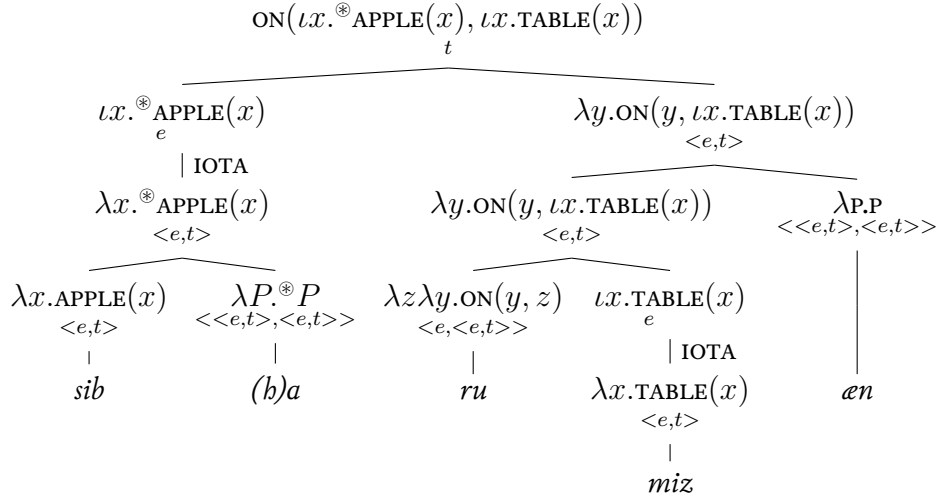


Figure 4: Example derivation tree for the plural definite construction

(43). On the other hand, we have seen examples that show bare plurals can sometimes have indefinite interpretations and we have also seen many examples of indefinite plurals with the clitic *-i*. If we build definiteness into the semantics of the plural morpheme we will rule out the incorrect indefinite interpretations of plurals but we will also rule out the correct ones. I think this dilemma points to the nature of definiteness in Persian: that instead of being lexically encoded, it is derived more globally and at the level of constructions and utterances. The fact that QUD and tense/aspect can also affect it also points in this direction. Therefore when it comes to definiteness, I prefer to leave the semantic system flexible for now and add required restrictions elsewhere. What exactly rules out an indefinite reading for (43) and similar constructions remains unclear for now.

Next I consider the singular indefinite construction with the numeral *je(k)* in example (44). The main difference between the bare nominal indefinite and the singular indefinite is that the singular indefinite cannot apply to parts or portions anymore. Example (44) is true and felicitous in Scenario 1 of Figure 1, technically true but not felicitous in Scenario 2, false for Scenario 3 and 4. Uttering a singular indefinite like (44) in Scenario 2 is similar to the English counterpart ‘an apple’ where the statement is considered under-informative and possibly false. One can try to make more sense of Scenario 3, if there was an apple that was sliced and the speaker is referring to the former state. However, this seems like adding assumptions to rescue the infelicity or falsehood of such an utterance. In short, the basic insight is that introduction of the numeral *je(k)* limits the description to singular apples rather than apple portions or apple sums.

- (44) *je sib ru miz e*
 one apple on table be.3.SG
 ‘There is an apple on the table’

Unlike Jasbi (2016, 2020a), I do not take *je(k)* to be an indefinite determiner like the English *a(n)*, introducing an existential quantifier. Such an analysis seems incorrect, given the fact that *je(k)* can combine with demonstrative determiners like *in* ‘this’ and *un* ‘that’ to create *in je sib* ‘this one apple’. The English *a(n)*, on the other hand, cannot combine with determiners like *this* or *that*. These observations suggest that Jasbi (2016, 2020a)’s analysis of *je(k)* was likely mistaken and that *je(k)* is a numeral like

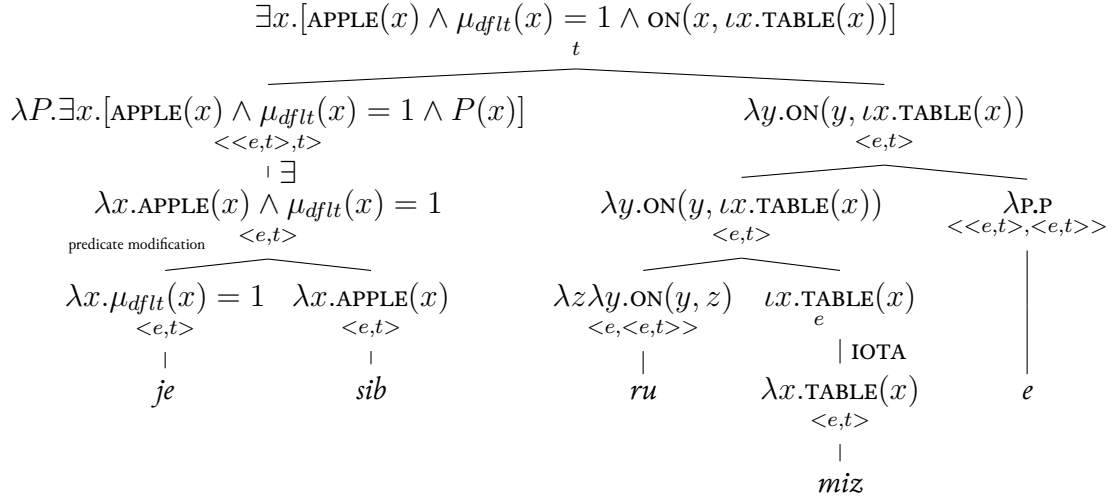


Figure 5: Example derivation tree for the simple singular indefinite construction

“one” and not a determiner. When left without a demonstrative determiner, the nominal along with the numeral modifier can be type-shifted to have existential force similar to what I had with bare nominals that receive existential reading in (41). Figure 5 shows the semantic derivation for example (44). The numeral *je(k)* ‘one’ in this analysis simply asserts that some objects have the cardinality 1 when measured by a default measure function μ_{dflt} . The default measure function is one that uses the common or “natural” unit of measurement. In case of apples it picks the natural unit of an individual apple rather than possible alternative measures like kilos, tons, slices, or baskets. In the analysis provided in Figure 5, the numeral directly combines with the predicate *sib* ‘apple’ using the Predicate Modification rule that typically allows conjunctive composition of two $\langle e, t \rangle$ adjectives and nouns. Once the combination of the numeral and the noun delivers singular nouns, we can covertly type-shift the noun phrase and have existential quantification over singular entities (e.g. apples). This analysis captures the observation that example (44) is true in scenarios 1 and 2 (although not optimal for 2), but false in scenarios 3 and 4.

- (45) *do/se/.../ʃænd ta sib ru miz e*
two/three/.../NUM CL apple on table be.3.PL
‘There are two apples on the table’

Finally, I provide an analysis of the classifier plural construction like the one in example (45). Since the exact number does not matter for the analysis, I use slashes to suggest that you can use any number word like *do* ‘two’, *se* ‘three’, or the unknown numeral word *ʃænd*. The main idea here is that the numeral provides the number and the classifier provides the measure function as Figure 6 shows. Similar to Scontras (2014), I treat individual numbers as entities in the model. I treat the unknown numeral as an individual variable ranging over numbers. This treatment will allow us to have the appropriate semantics for examples of the unknown numeral in questions as well, where it asks about cardinality and can be answered with any number word. Once the numeral and the classifier compose, the rest of the composition proceeds as we saw previously with the numeral ‘one’. Persian varieties that use the numeral ‘one’ with the classifier *ta* can be analyzed to have a uniform treatment of all numbers rather than separating ‘one’ from other numbers and treating it differently morphosyntactically. The analysis provided in Figure 6 predicts correctly that example (45) with the number three or the unknown numeral

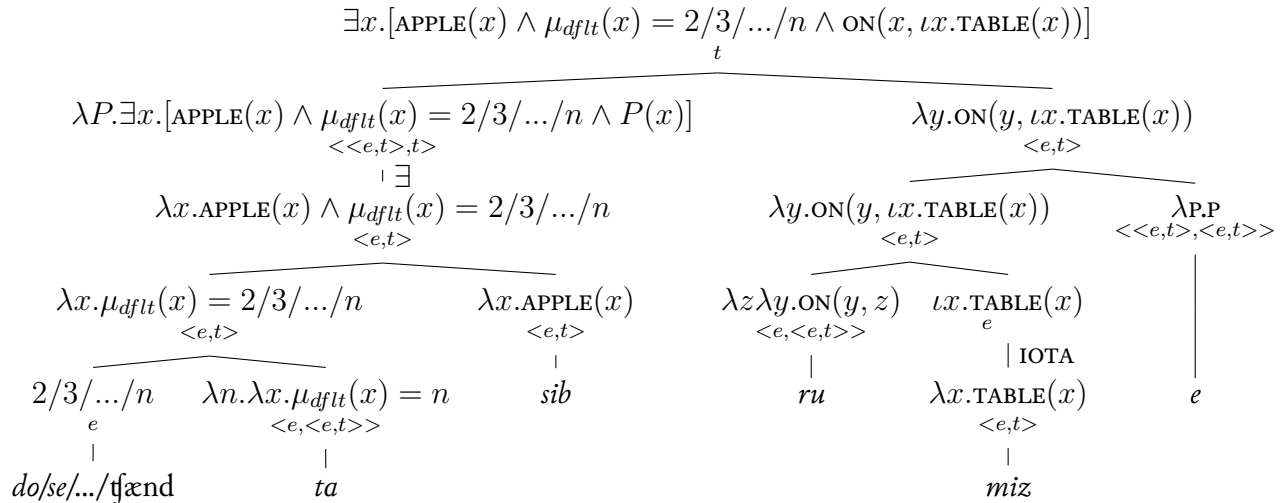


Figure 6: Example derivation tree for the plural indefinite construction

will be true in Scenario 2 but not the other scenarios.

The above analysis also predicts that the indefinite determiner and the plural marker cannot co-occur given that they encode contradictory meanings. This prediction seems to be borne out as example (46) shows. However, as explained before the plural marker can appear with the indefinite determiner if the indefinite clitic is also present. The analysis provided in this paper will rule out such constructions incorrectly. What does the indefinite clitic add that resolves this contradiction and allows for the composition of all these morphemes? It is not clear to me at this point so I leave this issue for future research.

- (46) a. **je sib-a ru miz æn*
IND apple-PL on table be.3.PL
'Some apples are on the table.'
- b. *je sib-a-i ru miz æn*
IND apple-PL-IND on table be.3.PL
'Some apples are on the table.'

5 Conclusion

The main claim of this paper is that plurality and definiteness are introduced via separate mechanisms: plurality is introduced morphologically via the plural marker or numeral classifiers, and (in)definiteness is introduced via covert operations that can be cued through different factors such as prosody, world-knowledge, or tense and aspect. The paper discussed five different plural constructions: bare plurals, indefinite plurals, classifier plurals, negative plurals, and constituent question plurals. Widespread indefinite interpretations of plurals in bare plural and indefinite plural constructions suggest that the plural morpheme cannot entail definiteness. On the other hand, persistent plural interpretations in constituent question plurals and unacceptability of the plural morpheme *-ha* in downward-entailing and non-veridical environments suggest that Persian plurals may denote plural entities exclusively and

not in addition to singulars unlike English. Whether in singulars or plurals, definiteness and indefiniteness is the end result of covert operations that are triggered by the presence or absence of several factors including numeral morphemes, the indefinite clitic, intonation, case marking, specificity marking, tense and aspect. I presented a compositional analysis in which bare nouns refer to singular, plural, and potentially portions of entities. The plural morpheme *-ha* simply restricts the nominal meaning to refer to plural entities. The definiteness of the nominal is determined in the process of composition and covertly rather than being encoded in any specific morpheme.

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Abbreviations

PL	Plural	IMP	Imperfective	NUM	(Unknown) Numeral
SG	Singular	EZ	Ezafe	SUB	Subjunctive
IND	Indefinite	PERF	Perfective	NEG	Negation
1	First Person	NPST	Non-Past	CL	Classifier
2	Second Person	ACC	Accusative	PST	Past
3	Third Person	NOM	Nominative		

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